Updated: 10 Aug 2026

José Federico Castro Tramontina, MS, BS

Master of Science in Cyber Defense (FIE)

Master of Science in Cryptography and Network Security (FIE)

CONTACT INFORMATION

Argentina – Buenos Aires
E-mail
WhatsApp
GitHub
Google Scholar
LinkedIn
ORCID
ResearchGate
Telegram
X

ABOUT

Army Officer (Captain), Cryptographer, Software and Embedded Systems Developer. Electronics Engineering Scholarship Student and PhD Candidate in Computer Science, with research interests in Cryptography, Industrial Cybersecurity, Control Engineering and Data Networks. Currently working on the development of a post-quantum key exchange protocol and the deployment of local AI models for various purposes.

EDUCATION

Doctor of Philosophy – PhD Candidate in Computer Science August 2020 – Present

Universidad Abierta Interamericana – Facultad de Tecnología Informática

Dissertation: Design, Development, and Cryptanalysis of a Key Exchange Protocol Based on Cayley-Dickson Algebras

Advisor: Juan Pedro Hecht, PhD, BS

Supervisor: Jorge Alejandro Kamlofsky, MS, BS

City of Buenos Aires, Argentina

Master of Science – MS Student in Data Networks August 2021 – Present

Universidad Tecnológica Nacional – Facultad Regional Mendoza

Thesis Proposal: Design and Implementation of a Secure Embedded Systems Network Based on Post-Quantum Cryptography

Advisor: Jorge Alejandro Kamlofsky, MS, BS

Mendoza, Argentina

Master of Science – MS in Cyber Defense March 2020 – March 2022

Universidad de la Defensa Nacional – Facultad de Ingeniería del Ejército

Thesis: Quantum Computing and the Breakdown of Contemporary Cryptography in Argentine Army Combat Units

Advisor: Alejandro Luis Echazú, MS, BS

City of Buenos Aires, Argentina

Master of Science – MS in Cryptography and Network Security March 2020 – May 2021

Universidad de la Defensa Nacional – Facultad de Ingeniería del Ejército

Thesis: Quantum Computing and Contemporary Cryptography – Preparing for the Future Breakdown of Current Cryptographic Algorithms

Advisor: Marcelo José Cipriano, MS, BS

City of Buenos Aires, Argentina

Bachelor of Science – BS in Operations Management March 2010 – December 2013

Universidad de la Defensa Nacional – Facultad del Ejército (Colegio Militar de la Nación)

El Palomar – Buenos Aires, Argentina

Bachelor of Science – BS Student in Electronics Engineering February 2023 – Present

Universidad de la Defensa Nacional – Facultad de Ingeniería del Ejército

City of Buenos Aires, Argentina

PROGRAMMING LANGUAGES, TECHNOLOGIES AND TOOLS

Programming Languages and Web Development

- Bash
- C
- C++
- CSS
- Google Apps Script
- HTML5
- JavaScript
- Python
- SQL

Backend, Data, CLI and Version Control

- FastAPI
- Git
- GitHub
- Rich
- SQLite
- Typer
- WebSockets

Artificial Intelligence and Local Models

- Antigravity
- Comfy
- ChatGPT
- Claude
- Codex
- DeepSeek
- Gemini
- Llama.cpp
- LM Studio
- OpenClaw

Computational Mathematics, Simulation, and Analysis

- CoCalc
- Jupyter Notebook / Google Colab
- LTSpice
- MATLAB / GNU Octave
- Microsoft Excel / LibreOffice Calc / Google Sheets
- Multisim
- SageMath

Systems, Virtualization, Administration, and Software Quality

- Crontab
- Docker
- GNU/Linux (Debian, Parrot Security, Ubuntu Server, etc.)
- Hypothesis
- MkDocs
- Moodle
- mypy
- Podman
- pytest
- Rclone
- Ruff
- Termux
- uv
- VirtualBox

Networking and Cybersecurity

- Ettercap
- Nmap
- OpenSSH
- Samba
- tcpdump
- Wireshark

Embedded Systems and Electronics

- adb (Android Debug Bridge)
- Arduino
- KiCad
- ESP32
- STM32
- Raspberry Pi Pico
- Raspberry Pi

Industrial Automation and Control

- 4diac (IDE and FORTE)
- EcoStruxure ControlExpert
- EcoStruxure MachineExpert
- Ladder – PLC Programming
- OpenPLC
- ScadaBR
- ST (Structured Text) – PLC Programming
- TIA Portal

PARTICIPATION IN RESEARCH, DEVELOPMENT AND TECHNOLOGICAL INNOVATION PROJECTS

Cybersecurity in Precision Agriculture and Livestock Farming in Argentina

Universidad Abierta Interamericana – Centro de Altos Estudios en Tecnología Informática (CAETI)

Objective: To propose Precision Agriculture and Livestock Farming processes and technological solutions adapted to the Argentine agricultural sector, incorporating cybersecurity measures and cryptographic solutions that ensure confidentiality and integrity without compromising system availability, while achieving measurable improvements in production efficiency.

Director: Jorge Alejandro Kamlofsky, MS, BS

Co-Director: José Federico Castro Tramontina, MS, BS

Ongoing developments:

- Systematic Literature Review

Cybersecurity in Industrial Control Systems

Universidad Abierta Interamericana – Centro de Altos Estudios en Tecnología Informática (CAETI)

Objective: To analyze and develop security recommendations and solutions based on improvements in network processes, technologies, and topologies. Research focuses on developing solutions that enhance confidentiality and integrity through Cryptography and Artificial Intelligence, with innovative applications in both local and international industry.

Director: Jorge Alejandro Kamlofsky, MS, BS

Ongoing developments:

- Design of an industrial network using current industry-standard tools
- Analysis and exploitation of vulnerabilities in communications based on the Modbus TCP/IP protocol
- Development of scripts for modifying PLC behavior
- Development of scripts for system event monitoring and automation of routine tasks
- PLC programming in Ladder language for AC motor control
- Update and improvement of the HK17 key exchange protocol

PARTICIPATION IN CONFERENCES AND CONGRESSES

- ECEFI 2022 – Séptimo Congreso Internacional de Educadores en Ciencias Empíricas en Facultades de Ingeniería
- JAIIO 52 – II Simposio Argentino de Ciberseguridad
- Ekoparty 2025 – Ciberseguridad en Ambientes Industriales: Desafíos y Aportes (Criptografía en ICS)

PUBLICATIONS

[1] J. F. Castro Tramontina, S. Pérez, and H. Facchini, “Análisis de Tiempo Real de Tráfico de Redes de Datos mediante Técnicas de Inteligencia Artificial,” ECEFI 2022 - Séptimo Congreso Internacional de Educadores en Ciencias Empíricas en Facultades de Ingeniería, vol. 7, 2022, Accessed: Nov. 27, 2022. [Online]. Available: https://www.researchgate.net/publication/362830143_Analisis_de_Tiempo_Real_de_Trafico_de_Redes_de_Datos_mediante_Tecnicas_de_Inteligencia_Artificial

[2] J. F. Castro Tramontina, C. Neil, J. Kamlofsky, P. Hecht, “Criptografía aplicada en entornos industriales: un mapeo sistemático de la literatura”, II Simposio Argentino de Ciberseguridad, 2023, pp. 58-73. Accessed: Oct. 19, 2023. [Online]. Available: https://ojs.sadio.org.ar/index.php/JAIIO/article/view/606/532

[3] J. Kamlofsky, J. F. Castro Tramontina, D. Manrique, J. Schmidt, “Ciberseguridad en Redes Operacionales”, Workshop de Investigadores en Ciencias de la Computación, 2026. Accessed: May 08, 2026. [Online]. Available: https://wicc2026.uai.edu.ar/posters/seguridad%20informatica/16373.pdf

CERTIFICATIONS, COURSES AND TRAINING

- Instructor in Tactical Radio Communication Networks (HARRIS and ELBIT Systems) – Ejército Argentino

AWARDS, DISTINCTIONS AND RECOGNITIONS

- Infantry Branch Merit Award – Highest GPA in the Basic Staff Officer Course (Sep 15, 2023)

MASTER’S THESIS EXAMINING COMMITTEE

Impact of Artificial Intelligence and Blockchain on Enhancing Security in FinTech Platforms **May 2026**

Universidad Abierta Interamericana – Facultad de Tecnología Informática

Candidate: Gabriel Ibarra, BS

Advisor: Francisco Gindre, MS, BS

City of Buenos Aires, Argentina

LANGUAGES

- Spanish: Native
- English: Level 6 – Escuela de Idiomas del Ejército (B2 – MCER)

WORK EXPERIENCE IN THE PRIVATE SECTOR

Software and Embedded Systems Developer **May 2026 – Present**

Freelance – City of Buenos Aires

- Farmacia Vital Olivos : Development of a digital queue management system for a pharmacy, based on a real-time client-server architecture, including:
  - Touchscreen kiosk for customers
  - Call management panel for cashiers/operators
  - Queue display screen (current and previous tickets) with integrated digital multimedia signage (images/videos)

Implemented technologies:

- Python
- FastAPI
- WebSockets
- SQLite
- HTML5
- CSS3
- JavaScript

Cybersecurity Consultant **November 2024 – Present**

Freelance – City of Buenos Aires

- genomIT : Advisory services on regulatory compliance, personal data protection, and strengthening security policies in accordance with ISO 27001

WORK EXPERIENCE IN THE ARGENTINE ARMY

Electronics Engineering Scholarship Student **February 2023 – Present**

Army Engineering College – City of Buenos Aires

- Full-time Electronics Engineering scholarship student with exclusive academic dedication

Chief Online Learning Officer – Chief Technology Officer – Military Professor **February 2021 – December 2022**

Army Infantry School – Campo de Mayo, Buenos Aires

- Systems administrator
- Automation of routine tasks using Bash scripting, Google Sheets, and Python
- Head of the online learning stages for all regular career courses conducted by the Infantry School
- Maintenance and administration of the “TC FERNÁNDEZ CUTIELLOS” Virtual Campus (Moodle)
- Military instructor for regular and complementary courses for Officers and NCOs

Language Student (German) **February 2020 – February 2021**

Defense Language Institute Foreign Language Center – City of Buenos Aires

- Preparation for an overseas assignment representing the Argentine Army (cancelled due to the COVID-19 pandemic)

Squad Commander – Communications Officer – Chief Information Officer **February 2018 – December 2019**

603rd Commandos Company – Bahía Blanca, Buenos Aires

- Responsible for the education and training of squad-level specialized personnel (up to 12 operators)
- Responsible for field and garrison radio communication links
- Responsible for the configuration and maintenance of IT and radio communication equipment

Platoon Leader – Counterintelligence Security Measures Officer **June 2014 – December 2017**

26th Mountain Infantry Regiment – Junín de los Andes, Neuquén

- Responsible for the education and training of platoon-level personnel (up to 40 members)
- Responsible for Information Security

TEACHING EXPERIENCE

Assistant Professor (Discrete Mathematics – Associate Degree in High School Mathematics Education) **March 2026 – August 2026**

Colegio Leonardo Da Vinci

- Computational Module:
  - Unit I: Graph Modeling
  - Unit II: Sequences and Recurrence Relations
  - Unit III: Counting Numbers and Combinatorics
  - Unit IV: The Language of Machines
  - Unit V: Fundamentals of Machine Operation and the Limits of Computability
- Specific Module:
  - Introduction to Algebraic Structures
  - Number Systems
  - First-Order Predicate Logic
  - Alternative Logical Systems

Assistant Professor (Discrete Mathematics – Bachelor of Science in Data Science Engineering | Bachelor of Science in Artificial Intelligence Engineering | Bachelor of Science in Cybersecurity | Bachelor of Science in Computer Science | Bachelor of Science in Artificial Intelligence | Bachelor of Science in Networks and Data Communications | Bachelor of Science in Information Technology) **March 2026 – August 2026**

Universidad de Palermo – Facultad de Ingeniería

- Unit I: Formal Languages
- Unit II: Finite-State Machines
- Unit III: Automata Theory
- Unit IV: Graph Theory
- Unit V: Integer Sets and Number Theory
- Unit VI: Algebraic Structures

Assistant Professor (Discrete Mathematics and Automata Theory – Bachelor of Science in Information Systems Engineering) **August 2025 – August 2026**

Universidad Abierta Interamericana – Facultad de Tecnología Informática (Sede Centro)

- Unit I: Graph Modeling
- Unit II: Sequences and Recurrence Relations
- Unit III: Counting Numbers and Combinatorics
- Unit IV: The Language of Machines
- Unit V: Fundamentals of Machine Operation and the Limits of Computability

Assistant Professor (Discrete Mathematics – Associate Degree in Information Systems) **March 2025 – August 2026**

Colegio Leonardo Da Vinci

- Unit I: Graph Modeling
- Unit II: Sequences and Recurrence Relations
- Unit III: Counting Numbers and Combinatorics
- Unit IV: The Language of Machines
- Unit V: Fundamentals of Machine Operation and the Limits of Computability

Assistant Professor (Physics – Associate Degree in Information Systems) **March 2025 – August 2026**

Colegio Leonardo Da Vinci

- Unit I: Units, Physical Quantities, Vectors, and Kinematics
- Unit II: Dynamics, Newton's Laws, Work, Energy, and Momentum
- Unit III: Thermodynamics and Waves
- Unit IV: Optics and Electrostatics
- Unit V: Introduction to Quantum Mechanics

Assistant Professor – Professor of Practice (Cryptography – Bachelor's Degree in Cybersecurity) **October 2023 – Present**

Universidad de la Fraternidad de Agrupaciones Santo Tomás de Aquino – Facultad de Ingeniería

- Unit I: Introduction to Cryptography
- Unit II: Mathematical Foundations of Cryptography
- Unit III: Applications of Modern Cryptography
- Unit IV: Emerging Trends and Applications

Associate Professor (Introduction to Computer Security – Associate Degree in Digital Forensics) **August 2024 – December 2024**

Instituto de Educación Superior ITEMED

- Unit I: Fundamentals of Information Security
- Unit II: Risk Management and Human Factors
- Unit III: Regulatory and Compliance Frameworks
- Unit IV: Access Control, Authentication, and Security in Operating Systems and Applications